

Exact Spatial Flatness and the Spin–Torsion Link: an Open Problem in Einstein–Cartan Cosmology

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Standard cosmology treats the observed spatial flatness of the Universe ($\Omega_K \simeq 0$) as an observational coincidence, to be explained dynamically by inflation. We adopt a different stance and state it as a postulate: spatial flatness is *exact*, $\Omega_K \equiv 0$, as a topological property of spacetime rather than a dynamical outcome. Unlike inflationary flattening, which predicts only $\Omega_K \approx 0$, this is a sharp and falsifiable claim: a measurement of $\Omega_K \neq 0$ at any precision would exclude it.

We are equally explicit about what this postulate does *not* do. Because $\sum_i \Omega_i = 1$ is an identity once $k = 0$ — the critical density $\rho_{\text{crit}} = 3H^2/8\pi G$ being recomputed with H — the closure condition cannot enforce any compensating dark-energy component, and constrains neither ρ_{DE} nor its equation of state. Earlier versions of this Letter presented such a compensation as a mechanism; that claim is withdrawn.

What Einstein–Cartan cosmology does supply, independently of this postulate, is a *static* dark-energy term derived from the action: the FLRW reduction of the axial torsion sector yields a background condensate with $w = -1$, its fluctuations behaving as radiation. The observed *evolution* of the equation of state does not follow, and remains calibrated.

This leaves a well-posed open problem, which we state rather than solve. The Weyssenhoff spin fluid that reduces the sound horizon before recombination, and the frozen condensate that sources dark energy, are two regimes of the *same* torsion field. A relation between Ω_{spin} and Ω_τ is therefore expected on physical grounds — but it must be derived from the action, not imposed by Ω -bookkeeping. We formulate this problem precisely, show why the route through the closure condition fails, and identify the propagating torsion mode as the direction in which a derivation could be sought.

Version note (v8). *This version withdraws the compensation mechanism presented in v4–v7 and reformulates the Letter around the open problem it leaves. The postulate is retained, with its scope restricted to exact spatial flatness. Supersedes preprint v5 (Zenodo, concept doi:10.5281/zenodo.19798923). Key changes: (i) the closure condition is shown to be an identity, incapable of enforcing a compensating debt; (ii) the static dark-energy term is attributed to the FLRW reduction of the action (Foundation II, Appendix J), independently of the postulate; (iii) the $\Omega_{\text{spin}}\text{--}\Omega_\tau$ relation is stated as the central open question rather than assumed; (iv) the analogy with charge conservation is withdrawn.*

Keywords: Einstein–Cartan cosmology; Topological Invariance Principle; dark energy; torsion; spatial flatness; cosmological constant; spin-torsion coupling; Weyssenhoff fluid; geometric gauge postulate; non-singular bounce; QCD phase transition; DESI; cosmic birefringence; Hubble tension; S_8 tension.

INTRODUCTION

Precision cosmology, largely driven by Cosmic Microwave Background (CMB) measurements from the Planck satellite [3], dictates that the observable Universe is remarkably flat, with the curvature parameter $\Omega_K = 0.0007 \pm 0.0019$. Within the standard Λ CDM

framework, this flatness is a dynamically unstable equilibrium, which primordial inflation resolves by driving Ω_K towards zero. Furthermore, the accelerated expansion of the Universe is attributed to a cosmological constant (Λ), representing approximately 70% of the total energy density today, yet lacking a fundamental microscopic origin. The observational pressure has recently intensified: the H0 Distance Network Collaboration has established $H_0 = 73.50 \pm 0.81 \text{ km s}^{-1} \text{ Mpc}^{-1}$ at $\sim 1\%$ precision [7], yielding a 7.1σ tension with flat Λ CDM (Planck+SPT+ACT) and a 5.0σ tension with BBN+BAO (DESI DR2). This makes the status of the flatness constraint, and of the geometric origin of Ω_Λ , worth stating precisely — which is the purpose of this Letter. The vacuum energy catastrophe—a discrepancy

of 10^{55} to 10^{122} orders of magnitude between QFT predictions and the observed Λ [10, 11]—remains unresolved; the exact-flatness postulate adopted here does not resolve it either, and we make no claim that it does.

In this Letter, we introduce a paradigm shift regarding the nature of spatial flatness and cosmic acceleration. Extending general relativity to the Einstein-Cartan (EC) framework—where spacetime torsion is effectively algebraic in the low-density limit (ECKS [4, 9]), and acquires propagating degrees of freedom via Poincaré Gauge Theory (PGT) at high density [2]—we propose that spatial flatness is not an accident of initial conditions, but an *exact* property of the extended geometric framework—posited, not enforced by any dynamical response.

The key phenomenological calibration—the sound-horizon reduction, and the single stiffness parameter that addresses both the H_0 and S_8 tensions—is presented in the companion paper Foundation I [1]; the microscopic dark sector and baryogenesis mechanism are developed in Foundation II [2]. This division of labor is deliberate: Foundation I provides the numerical calibration of (H_0, S_8, r_s) , Foundation II supplies the microscopic torsion condensate and baryogenesis sectors, and the present Letter isolates what the Topological Invariance Principle does and does not underwrite—exact spatial flatness, and not the dark-energy sector.

THE POSTULATE, AND ITS EXACT SCOPE

What the postulate does *not* do

We state at the outset what this postulate cannot deliver, because earlier versions of this Letter claimed otherwise.

Once $k = 0$, dividing the Friedmann equation by H^2 gives $\sum_i \Omega_i = 1$ *identically*, for any matter content whatsoever: adding a spin component changes H , and the critical density $\rho_{\text{crit}} = 3H^2/8\pi G$ follows. No deficit arises, hence nothing is compensated. The closure condition is an **identity**, not a constraint: it fixes neither ρ_{DE} nor its equation of state.

The point is sharpened by a comparison already made in Foundation II [2]. In pure Λ CDM, with ρ_Λ strictly constant ($w = -1$ exactly), Ω_Λ grows from 0.002 at $z = 10$ to 0.700 today, simply because $\rho_m \propto a^{-3}$ dilutes—an artefact of the denominator, not a strengthening of the dark sector. *A growing Ω is therefore not evidence of an evolving dark sector.* What would carry physical content is an evolving ρ , and the closure condition says nothing about that.

Withdrawn statements. Earlier versions of this Letter stated that any injected spin density algebraically enforces a compensating torsion debt, and that the torsion fraction increases mechanically to maintain topological equilibrium. Neither survives the argument above, and

both are withdrawn. We also withdraw the analogy with charge conservation: in FRW, k is a constant of the solution, not a Noether charge—there is nothing to conserve.

What it does do

What remains is a sharp and *falsifiable* claim: spatial flatness is **exact**, $\Omega_K \equiv 0$, as a topological property rather than a dynamical outcome. This is not what inflation delivers: inflationary flattening predicts $\Omega_K \approx 0$, typically at the 10^{-5} level. A measurement of $\Omega_K \neq 0$ at *any* precision would exclude the present framework, while leaving inflation untouched. That asymmetry is the whole empirical content of the postulate—and, we argue, enough to make it worth stating.

We formulate the *Topological Invariance Principle* (TIP) as follows: *Global spatial flatness ($\Omega_{\text{tot}} \equiv 1$) is an absolute and inviolable geometric equilibrium state, not a contingent initial condition. The same axial torsion field appears in two regimes across cosmic history—as a spin fluid at high density prior to recombination (Ω_{spin}), and as a frozen condensate at late times (Ω_τ)—whose scale-dependent structures are detailed in the companion papers [1, 2]. Whether a relation between the two follows from the action is the open problem posed in Sec. ; it is not supplied by the closure condition.*

Under this principle, the standard Friedmann equation for fractional densities is extended to include the spin-torsion pair:

$$\Omega_m + \Omega_r + \Omega_{\text{spin}} + \Omega_\tau + \Omega_K = 1, \quad \Omega_K \equiv 0 \quad \forall t \quad (1)$$

where Ω_τ is the macroscopic torsion density that, in the standard parameterisation, plays the role of Ω_Λ : at leading order it acts as an effective cosmological constant, while its full dynamical equation of state $w_\tau(z)$ departs from -1 at late times, yielding $w_0 = -0.904$, $w_a = -0.153$ [1].

The TIP raises $\Omega_K = 0$ from an observational coincidence to an exact topological property. It is *not* enforced dynamically by the pair $(\Omega_{\text{spin}}, \Omega_\tau)$: that these are two regimes of the same axial field is expected on physical grounds, but the relation between them is the open problem of Sec. , not a consequence of the postulate.

To preserve topological invariance against the massive injection of chiral spin during the early radiation era, the torsion density must perfectly neutralize the spin injection at the moment of peak amplitude:

$$\Omega_\tau^{(\text{initial})} = -\Omega_{\text{spin}}^{(\text{peak})} \quad (2)$$

The physical origin of $\Omega_{\text{spin}}^{(\text{peak})}$ is the fermion spin-density at the QCD transition ($T \sim 200$ MeV), when the quark-gluon plasma hadronises into a dense spin-carrying fermion gas whose collective spin scales as $\rho_{\text{spin}} \propto a^{-6}$. The TIP condition (2) uniquely fixes the initial torsion

debt from the physics of the QCD epoch, without any additional free parameter.

This condition is not imposed by hand; it is the unique value for which the right-hand side of the modified Friedmann equation

$$H^2 = \frac{8\pi G}{3}(\rho_r + \rho_m + \rho_\Lambda - \rho_{\text{spin}})$$

reaches zero at finite a_{min} — the non-singular bounce — while simultaneously satisfying $\Omega_{\text{tot}} = 1$ at all subsequent epochs. The numerical value of $\Omega_{\text{spin}}^{(\text{peak})}$ depends on the fermion spin density at the QCD epoch; its observational calibration from the H_0 , S_8 and BAO tensions is established in Foundation I [1], which obtains $\Omega_{\text{spin}}^{(\text{peak})} \approx 0.093$. Any future revision of that calibration would modify the numerical value entering Eq. (6), without altering the logical structure of the TIP itself. *Robustness note:* the TIP mechanism and its qualitative predictions (geometric dark energy, non-singular bounce, birefringence) hold for any value of $\Omega_{\text{spin}}^{(\text{peak})}$ consistent with QCD thermodynamics, i.e. broadly $\Omega_{\text{spin}}^{(\text{peak})} \in [0.05, 0.15]$. The precise value 0.093 is an observational calibration, not a logical prerequisite of the principle.

This initial condition fixes the magnitude of the topological debt created at the onset of the spin phase. During the active spin epoch, the continuous compensation constraint holds:

$$\Omega_{\text{spin}}(a) + \Omega_\tau(a) = 0 \quad (3)$$

This constraint is consistent with causality because Ω_τ is a *global* topological quantity—the frozen vacuum expectation value $\langle S_\mu \rangle = s_0 \delta_\mu^0$ [2]—not a local propagating field responding to local perturbations. Torsion is thus fundamentally defined as the geometric counter-force required to maintain a strictly flat Universe in the presence of extreme fermion spin densities. The fate of this debt once the spin fluid dilutes is the subject of the next section.

WHAT THE ACTION DOES GIVE, INDEPENDENTLY OF THE POSTULATE

Dark energy in this framework does *not* come from the closure condition. It comes from the action, and the derivation is independent of Sec. .

Reducing the axial torsion sector on an FLRW background (Foundation II [2], Appendix J; symbolically verified) gives three results. First, for $S_\mu = (s(t), 0, 0, 0)$ the field strength vanishes identically, $F_{\mu\nu}^{(S)} \equiv 0$: in a homogeneous universe this field *cannot* carry kinetic energy. Second, the spin source diluting as a^{-6} , the field settles at the minimum of its potential, $W'(X_0) = 0$. Third, at that minimum the stress tensor reduces to

$$T_{\mu\nu} = -W(X_0) g_{\mu\nu}, \quad (4)$$

exactly the form of a cosmological constant, whence

$$\rho = W(X_0), \quad p = -W(X_0), \quad w = -1. \quad (5)$$

The static part of dark energy is therefore **derived**, not posited — and it owes nothing to the postulate of Sec. .

Independent corroboration — and what is, and is not, new here. That a torsion condensate can behave as a cosmological constant is not a result of this framework alone: it has been obtained independently, by different routes, by Ivanov and Wellenzohn [14] and by Yun and Lee [15]. We regard this convergence as a strength rather than a loss: the result is not an artefact of our particular reduction. *What is proper to the present framework is not the isolated result, but its integration* — the same axial torsion field supplies the spin fluid that reduces the sound horizon before recombination and the frozen condensate that sources Λ at late times. Whether that integration can be derived rather than observed is precisely the problem stated in Sec. .

The fluctuations give radiation, not dark energy. Writing $S_\mu = s_0 \delta_\mu^0 + \delta S_\mu$, the spatial components are massless Goldstone modes whose action is conformally invariant Maxwell; their mode equation $\delta S_k'' + k^2 \delta S_k = 0$ carries no Hubble friction, so $\rho_{\delta S} \propto a^{-4}$ and $w = +1/3$. *The observed evolution of the equation of state does not follow from this sector either.* The pair (w_0, w_a) remains calibrated.

THE OPEN PROBLEM

We now state the question this Letter is written to pose.

The Weyssenhoff spin fluid that reduces the sound horizon before recombination (Foundation I [1]) and the frozen condensate that sources dark energy (Sec.) are not two ingredients. They are **the same axial torsion field in two regimes**: dynamically sourced at high spin density, frozen at the minimum of its potential once that source has diluted. A relation between Ω_{spin} and Ω_τ is therefore natural to expect on physical grounds.

What is *not* legitimate is to obtain it from Ω -bookkeeping. As Sec. shows, the closure condition cannot supply such a relation: it is an identity. The link must come from the action, or not at all.

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Does the Poincaré Gauge action imply a relation between the spin density Ω_{spin} that governs the pre-recombination sound horizon and the condensate density Ω_τ that sources late-time dark energy; and if so, with which sign?

We do not answer it here. We state it, show why the route through the closure condition fails, and indicate where a derivation might be sought.

Where a derivation might be sought. The obstacle is not a no-go theorem. The known no-go on crossing $w = -1$ with a single algebraically slaved degree of freedom [13] forbids a *crossing* which the data do not require: a non-crossing model already reproduces the calibrated (w_0, w_a) to within a few per cent. Moreover, the negative result of this section — that a diluting, algebraically slaved torsion does not carry late-time dark energy — has itself been reached independently, by a different route, by Izaurieta, Lepe and Quinzacara [16]. What is needed is a torsion mode with a genuine kinetic term — a *propagating* axial mode, such as quadratic Riemann–Cartan terms are known to supply [12], and along which other groups are actively working. Deriving its potential from the action, rather than positing it, is the outstanding task. Ghost-freedom and Hamiltonian stability would then have to be checked.

This strict geometric reaction provides a natural, first-principles resolution to the nature of Dark Energy. The TIP mechanism operates in three stages: (i) a transient primordial spin density $\Omega_{\text{spin}} \propto a^{-6}$, sourced by fermion spin at the QCD confinement transition ($T \sim 200$ MeV), is observationally calibrated at the acoustic-era reference redshift $z \approx 7500$, where it reaches its cosmologically relevant amplitude $\Omega_{\text{spin}}^{(\text{peak})} \approx 0.093$ (Foundation I); (ii) as the spin fluid subsequently dilutes ($\rho_{\text{spin}} \propto a^{-6} \rightarrow 0$), the axial field settles at the minimum of its potential and freezes into the vacuum structure, persisting as a residual geometric density that plays the observational role of Λ (Sec.); (iii) the numerical relation $\Omega_{\tau}^{(\text{initial})} = -\Omega_{\text{spin}}^{(\text{peak})}$ used in the companion papers is an *empirical* identification, not a consequence of the closure condition — establishing whether the action implies such a relation, and with which sign, is the open problem of Sec. .

According to the TIP (Eq. 2), this transient primordial spin phase obligatorily generated a global topological debt:

$$\Omega_{\tau}^{(\text{initial})} \approx -0.093 \quad (6)$$

within the spacetime manifold. The sign is negative by construction of the TIP: Ω_{τ} must exactly cancel $\Omega_{\text{spin}}^{(\text{peak})}$ to maintain $\Omega_K \equiv 0$ (Eq. 3). The numerical magnitude $\Omega_{\text{spin}}^{(\text{peak})} \approx 0.093$ is set by the QCD transition physics and calibrated observationally in Foundation I [1]; it is reported here as an input, not re-derived. Crucially, while the kinetic spin fluid dilutes rapidly as a^{-6} due to cosmic expansion, the corresponding macroscopic torsion field (the geometric debt) undergoes a phase transition, “freezing” into the elastic structure of the vacuum once the spin source vanishes.

Formally, within the PGT extension of the ECKS action detailed in Foundation II [2], this freezing corresponds to the axial torsion pseudo-vector $S_{\mu} \equiv \frac{1}{6}\varepsilon_{\mu\nu\lambda\rho}T^{\nu\lambda\rho}$ acquiring a non-zero vacuum expectation

value,

$$\langle S_{\mu} \rangle = s_0 \delta_{\mu}^0 \neq 0, \quad (7)$$

whose energy density $\rho_{\Lambda}^{\text{eff}} = \frac{1}{2}m_T^2 s_0^2$ constitutes the effective cosmological constant at leading order. Its field-theoretic derivation—with the overall scale calibrated on the TIP debt—is developed in Foundation II [2]. At leading phenomenological order, this frozen torsion acts as an effective cosmological constant ($w_{\tau} = -1$). The full dynamical equation of state—an inherently time-varying $w(z)$ derived from the ECKS action—departs from this limit at late times, yielding $w_0 \approx -0.904$, $w_a \approx -0.153$. These values were derived *prior* to the DESI results from the PIT calibration [1] and are qualitatively consistent with the DESI BAO preference for $w \neq -1$ [5, 6]; a full likelihood comparison is deferred to Foundation III. The values $w_0 = -0.904$ and $w_a = -0.153$ are therefore not free fit parameters of the Letter itself; they are inherited from the Foundation I calibration and serve here as the late-time phenomenological imprint of the same geometric torsion sector. As the Universe expands, the absolute densities of baryonic matter and topological dark matter [2] dilute as a^{-3} . Consequently, their fractional sum (Ω_m) decreases. Because the fractional densities sum to unity by construction once $k = 0$, the residual 1 to satisfy the TIP (Eq. 1), the fractional weight of the frozen residual torsion (Ω_{τ}) is *mechanically* forced to increase over cosmic time to compensate for matter dilution (see Figure 1).

What the standard Λ CDM model interprets as a mysterious, dominant Dark Energy fluid ($\Omega_{\Lambda} \approx 0.7$) driving accelerated expansion is, in the ECF model, simply the geometric response of spacetime taking up a larger fraction of the cosmic budget to prevent the Universe from curving upon matter dilution.

CONCLUSION

It must be emphasised that the TIP operates here as a foundational *postulate* of an extended Einstein–Cartan framework, not as a theorem of the minimal ECKS Lagrangian. What Foundation II derives from the Poincaré Gauge Theory action is the torsion condensate that supplies a static dark-energy term via spontaneous symmetry breaking; the closure constraint $\Omega_{\text{total}} = 1$ itself remains a postulate, and its first-principles derivation—together with the $\Omega_{\text{spin}}\text{--}\Omega_{\tau}$ relation and its sign—is an open problem [2].

We close by separating what this Letter asserts, what it withdraws, and what it leaves open.

Asserted. Spatial flatness is exact, $\Omega_K \equiv 0$, by topological necessity rather than by dynamical relaxation. This is a falsifiable claim, and it is the whole content of the postulate: a measurement of $\Omega_K \neq 0$ at any precision

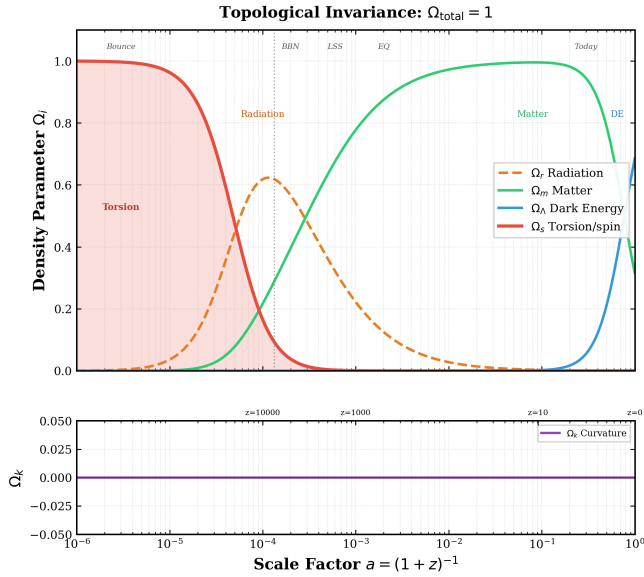


FIG. 1. **Topological Invariance.** Temporal evolution of fractional energy densities $\Omega_i(z)$ in the ECF model, computed with $F_{\text{ion}} = 1.2765$, $\Omega_{\text{spin,peak}} = 0.093$ [1]. The invariant $\Omega_{\text{tot}} = 1$ (bottom line) is the visual expression of the TIP. The late-time rise of Ω_Λ is an arithmetical consequence of matter dilution — it occurs for a strict cosmological constant too — not the signature of a new field. Reproducible via `plot_Cosmic_History_Omegas.py` (public repo).

would exclude the framework while leaving inflation untouched.

Withdrawn. The claim, present in earlier versions, that the closure condition enforces a compensating torsion debt and thereby generates dark energy. It does not: $\sum_i \Omega_i = 1$ is an identity once $k = 0$, and a growing Ω is equally a feature of a strict cosmological constant.

Left open. Whether the Poincaré Gauge action implies a relation between Ω_{spin} and Ω_τ — the two regimes of a single axial torsion field — and with which sign. We regard this as the substantive question the framework now faces, and have stated it rather than assumed an answer. The static part of dark energy is *derived* from the FLRW reduction of the action (Sec.); its evolution remains calibrated, and relating it to the spin sector is the open problem of Sec. . At leading order the frozen torsion mimics Λ ; the calibrated $w(z)$ departs from it, in the direction DESI has since measured, though a full-covariance comparison places the point at 2.22σ from the DR1 central value and does not favour it over Λ CDM on those data alone (Foundation III). The detailed numerical work—the calibration addressing the H_0 and S_8 tensions (Foundation I [1]) and the modelling of topological dark matter via torsion knots (Foundation II [2])—is presented in the companion regular articles. A geometric consistency check of the frozen torsion debt yields a cosmic birefringence angle $\beta_{\text{ECF}} \approx 0.35^\circ$ (Planck/WMAP baseline), consistent with

the ACT-DR6 result $0.215 \pm 0.074^\circ$ at the 2.9σ level [8]. This is a calibrated consistency check, not a parameter-free prediction, and will be sharpened by LiteBIRD [1] (Sec. 7 of Foundation I).

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